

DRAGOPAINT

DRAGOCRYL VELOUTE SUP AIRLESS

Protect and decorate 1m² of substrate for a lifespan of 10 years.

Issued 30.03.2021
Valid until 30.03.2026

Third party verified
Conform to EN 15804+A2, NBN/DTD B08-001
and ISO 14025

Modules declared					
A123	A4	A5	B	C	D
•	•	•	•	•	•

[B-EPD n° 21-0127-012-00-00-EN]



OWNER OF THIS ENVIRONMENTAL PRODUCT DECLARATION
Dragopaint

EPD PROGRAM OPERATOR
**Federal Public Service of Health, Food Chain Safety
and Environment**
www.b-epd.be

PRODUCT DESCRIPTION

PRODUCT NAME

Commercial references: Dragocryl Velouté sup airless, and Peinture airless: velouté sup.

PRODUCT DESCRIPTION AND INTENDED USE

Dragocryl velouté sup airless is a water-based paint. The product has 1.41 kg/L density and an average coverage 8.55 m²/L, with an opacity of 95.48 %.

This is a specific EPD from a single company.

Dragocryl velouté sup airless is applied to interior walls and ceilings, following the standard EN 13300.

REFERENCE FLOW / FUNCTIONAL UNIT

The functional unit is: Protect and decorate 1 m² of substrate for a lifespan of 10 years, when applied according to DTU 59.1.

Packaging is included and consists of interlayer, plastic film, buckets and lids in polypropylene, and wooden pallets. This B-EPD results represent 15-liter buckets packaging.

The product weight per reference flow is 0.16 kg.
The density of the product is 1.41 kg / liter.

INSTALLATION

The scope of this B-EPD is as installed.

Product losses during application is accounted as 2%.

Materials for paint application are included. This EPD includes the impacts of all processes, and application materials necessary for applying the product.

The materials necessary for application are detailed in table below.

Application	Generic data	Quantity (kWh/FU)
Airless machine	Electricity, low voltage {BE} market for Cut-off, U	1.2E-3

IMAGES OF THE PRODUCT AND ITS INSTALLATION



Figure 1: Commercial references



Figure 2: Paint application by airless machine

COMPOSITION AND CONTENT

This paragraph is split up in the product, application materials and packaging compositions.

Components	Composition / content / ingredients	Quantity
Product	<i>Water</i> <i>Titanium dioxide</i> <i>Calcium carbonate</i> <i>Additives</i>	<i>15 - 20%</i> <i>8 – 13%</i> <i>28 - 33%</i> <i>40 - 45%</i>
Application materials	<i>Airless machine</i>	<i>1.2E-3 kWh/FU</i>
Packaging	<i>Interlayer (cardboard)</i> <i>Plastic film (polyethylene)</i> <i>Bucket (polypropylene)</i> <i>Lids (polypropylene)</i> <i>Wooden pallets</i>	<i>1.60E-4 kg/FU</i> <i>8.73E-5 kg/FU</i> <i>3.75E-3 kg/FU</i> <i>7.06E-4 kg/FU</i> <i>5.91E-3 kg/FU</i>

The product does not contain materials listed in the “Candidate list of Substances of Very High Concern for authorization”.

REFERENCE SERVICE LIFE

The reference service life is estimated at 10 years, according to TOTEM materials lifespan.

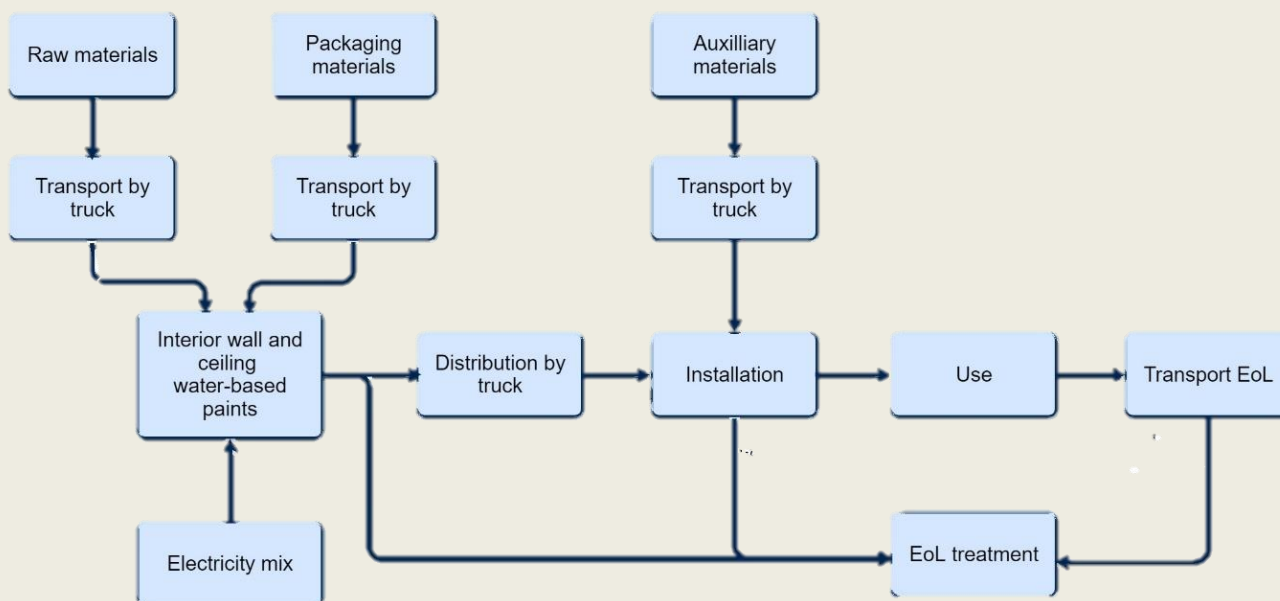
The conditions under which this RSL is valid are as following: if applied correctly, according to the manufacturer (following the DTU 59.1), Dragocryl Velouté sup airless does not need maintenance, repair, replacement, or refurbishment during the lifespan of 10 years.

DESCRIPTION OF GEOGRAPHICAL REPRESENTATIVITY

This EPD is a cradle-to-grave, and includes modules A123, A4, A5, B, C and D for the Belgian scenario. The EPD is representative for the Belgian market.

DESCRIPTION OF THE PRODUCTION PROCESS AND TECHNOLOGY

Raw materials are transported by truck to Dragopaint site in Belgium. The manufacturing process consist in mixing the ingredients, Belgian electricity mix is used in this purpose. Production losses represent 2.00 % in mass. Those losses are landfilled as inert waste in their end of life. The final product is packed in 15 liters buckets, interlayers, and wooden pallets, surrounded by plastic film. The packed product is then distributed by truck.



TECHNICAL DATA / PHYSICAL CHARACTERISTICS

Technical property	Standard	Value	Unit	Comment
Opacity		95.48	%	
Average coverage		8.55	m ² /liter	
Lifespan	DTU 59.1	10	years	
COV		0.467	g/liter	
Biocide		0.15	% m/m	
Paint density		1.41	kg/liter	

LCA STUDY

DATE OF LCA STUDY

Data used in this cradle-to-grave LCA study represent the year 2019.

SOFTWARE

For the calculation of the LCA results, the software program Simapro version 9.1.1 has been used.

INFORMATION ON ALLOCATION

No co-product allocation occurs in the product foreground system.

No multi-input allocation occurs in the product system.

The allocations from the background database are kept intact.

INFORMATION ON CUT OFF

Dragopaint provided the data used in this study. Some plausibility and completeness assessments and checks were conducted for some inputs. For a few remaining data, no extended assessment was conducted, therefore accepting data gaps.

In all cases, it is assumed that the cut-off criteria of EN 15804 are met.

INFORMATION ON EXCLUDED PROCESSES

Following processes were excluded for the inventory:

- The effects of capital goods and infrastructural processes have been excluded.
- Flows related to human activities such as employee transport and administration activities are excluded.
- Packaging of raw materials are excluded.

INFORMATION ON BIOGENIC CARBON MODELLING

Dragocryl Velouté sup airless water-based paint does not include any biogenic carbon content. The biomass present in the model corresponds to the packaging: wooden pallets.

For EN 15804+A2 include following table:

Biogenic carbon content (kg C / FU)	
Biogenic carbon content in product (at the gate)	
Biogenic carbon content in accompanying packaging (at the gate)	2.30E-3

INFORMATION ON CARBON OFFSETTING

Carbon offsetting is not allowed in the EN 15804 and hence not taken into account in the calculations.

ADDITIONAL OR DEVIATING CHARACTERISATION FACTORS

For EN 15804+A2: The characterization factors from EC-JRC were applied. No additional or deviating characterisation factors were used.

DESCRIPTION OF THE VARIABILITY

Not applicable. This is not an average EPD.

DATA

SPECIFICITY

The data used for the LCA are specific for this product which is manufactured by a single manufacturer in a single production site (Dragopaint).

PERIOD OF DATA COLLECTION

Manufacturer specific data have been collected for the year 2019.

INFORMATION ON DATA COLLECTION

Most of the generic data used in this B-EPD is from Ecoinvent 3.6. For some missing data, CEPE LCDN node: Chemicals for paints datasets for PEF calculations database were use.

Some adaptations in Ecoinvent 3.6 datasets were done to better adapt the primary data.

DATABASE USED FOR BACKGROUND DATA

Ecoinvent version 3.6 was used, released in September 2019, as well as CEPE LCDN node: Chemicals for paints datasets for PEF calculations database.

ENERGY MIX

The Belgian energy mix is considered for the manufacturing of the product. Therefore, the Belgian energy mix was used to declare the benefits beyond the system boundaries (for module D).

PRODUCTION SITE

Dragopaint, Rue de l'Europe, 42 (Zone industrielle), B-7080 Frameries, Belgium

SYSTEM BOUNDARIES

Product stage			Construction installation stage		Use stage							End of life stage				Beyond the system boundaries
Raw materials	Transport	Manufacturing	Transport	Construction installation stage	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery- Recycling-potential
A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒

X = included in the EPD
MND = module not declared

POTENTIAL ENVIRONMENTAL IMPACTS PER REFERENCE FLOW

The results of the LCIA are calculated for Dragocryl Velouté sup airless. The results are provided for coverage of 1m² using Dragocryl Velouté sup airless.

		Production			Construction process stage		Use stage							End-of-life stage				D Reuse, recovery, recycling
		A1 Raw material	A2 Transport	A3 manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	
	GWP total (kg CO2 equiv/FU)	2.82E-01	9.70E-03	1.22E-02	3.54E-03	6.46E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	8.68E-04	2.53E-04	3.44E-05	5.38E-04
	GWP fossil (kg CO2 equiv/FU)	2.81E-01	9.69E-03	1.52E-02	3.54E-03	6.20E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	8.67E-04	2.54E-04	3.43E-05	1.76E-04
	GWP biogenic (kg CO2 equiv/FU)	3.57E-04	5.18E-06	-3.05E-03	1.81E-06	2.50E-03	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	4.62E-07	-1.12E-06	1.11E-07	3.62E-04
	GWP luluc (kg CO2 equiv/FU)	1.15E-04	3.40E-06	1.28E-05	1.43E-06	1.99E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.03E-07	3.19E-07	6.72E-09	2.23E-07
	ODP (kg CFC 11 equiv/FU)	1.66E-08	2.20E-09	1.15E-09	7.90E-10	1.15E-08	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.97E-10	3.97E-11	1.67E-11	5.70E-10
	AP (mol H+ equiv/FU)	1.31E-03	3.81E-05	5.83E-05	1.43E-05	2.29E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.54E-06	1.96E-06	3.34E-07	7.46E-06
	EP - freshwater (kg P equiv/FU)	7.31E-06	7.63E-08	4.81E-07	3.10E-08	5.47E-07	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	6.81E-09	8.29E-09	2.56E-10	-1.03E-08
	EP - marine (kg N equiv/FU)	2.18E-04	1.10E-05	9.86E-06	4.14E-06	6.43E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.05E-06	4.92E-07	1.25E-07	3.49E-06
	EP - terrestrial (mol N equiv/FU)	2.32E-03	1.21E-04	1.12E-04	4.58E-05	7.10E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.16E-05	6.54E-06	1.38E-06	4.01E-05
	POCP (kg NMVOC equiv/FU)	8.18E-04	3.78E-05	4.40E-05	1.41E-05	2.32E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.56E-06	1.56E-06	3.95E-07	8.88E-06

	ADP Elements (kg Sb equiv/FU)	7.39E-07	1.88E-08	2.38E-08	8.34E-09	1.10E-07	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.67E-09	3.25E-09	2.37E-11	6.27E-09
	ADP fossil fuels (MJ/FU)	5.96E+00	1.46E-01	4.68E-01	5.30E-02	8.69E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.31E-02	5.44E-03	1.11E-03	-8.08E-02
	WDP (m³ eq deprived /FU)	1.31E-01	4.08E-04	9.38E-03	1.58E-04	4.95E-03	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.64E-05	5.88E-05	3.35E-06	-1.50E-03

GWP total = total Global Warming Potential (Climate Change); GWP-luluc = Global Warming Potential (Climate Change) land use and land use change; ODP = Ozone Depletion Potential; AP = Acidification Potential for Soil and Water; EP = Eutrophication Potential; POCP = Photochemical Ozone Creation; ADPE = Abiotic Depletion Potential – Elements; ADPF = Abiotic Depletion Potential – Fossil Fuels; WDP = water use (Water (user) deprivation potential, deprivation-weighted water consumption)

RESOURCE USE

	Production			Construction process		Use stage							End-of-life stage				
	A1 Raw material	A2 Transport	A3 manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	
PERE (MJ/FU, net calorific value)	1.89E-01	2.07E-03	-6.05E-02	8.52E-04	5.07E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.84E-04	5.66E-04	1.70E-05	-7.29E-03
PERM (MJ/FU, net calorific value)	0.00E+00	0.00E+00	9.08E-02	0.00E+00	-3.56E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT (MJ/FU, net calorific value)	1.89E-01	2.07E-03	3.04E-02	8.52E-04	1.51E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.84E-04	5.66E-04	1.70E-05	-7.29E-03
PENRE (MJ/FU, net calorific value)	6.50E+00	1.47E-01	2.98E-01	5.36E-02	1.01E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.32E-02	5.89E-03	1.10E-03	-8.72E-02
PENRM (MJ/FU, net calorific value)	0.00E+00	0.00E+00	2.00E-01	0.00E+00	-1.20E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT (MJ/FU, net calorific value)	6.50E+00	1.47E-01	4.99E-01	5.36E-02	8.87E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.32E-02	5.89E-03	1.10E-03	-8.72E-02

SM (kg/FU)	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF (MJ/FU, net calorific value)	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF (MJ/FU, net calorific value)	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW (m³ water eq/FU)	2.84E-03	1.12E-05	1.85E-04	4.40E-06	1.19E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	9.99E-07	1.48E-06	1.30E-06	-2.80E-05

PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy resources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Net use of fresh water

WASTE CATEGORIES & OUTPUT FLOWS

	Production			Construction process stage		Use stage							End-of-life stage				
	A1 Raw material	A2 Transport	A3 manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	D Reuse, recovery, recycling
Hazardous waste disposed (kg/FU)	1.98E-02	9.42E-05	2.53E-04	3.78E-05	9.54E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	8.41E-06	2.81E-05	4.56E-07	3.98E-05
Non-hazardous waste disposed (kg/FU)	7.02E-02	7.70E-03	6.11E-03	2.55E-03	4.39E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	6.87E-04	1.89E-04	8.00E-03	3.22E-03
Radioactive waste disposed (kg/FU)	7.64E-06	9.97E-07	1.10E-06	3.59E-07	5.22E-06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	8.91E-08	4.28E-08	7.53E-09	-8.60E-08
Components for re-use (kg/FU)	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Materials for recycling (kg/FU)	0.00E+00	0.00E+00	0.00E+00	0.00E+00	4.08E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Materials for energy recovery (kg/FU)	0.00E+00	0.00E+00	0.00E+00	0.00E+00	4.09E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported energy heat (MJ/FU)	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.57E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported energy electricity (MJ/FU)	0.00E+00	0.00E+00	0.00E+00	0.00E+00	7.96E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

IMPACT CATEGORIES ADDITIONAL TO EN 15804

		Production			Construction process		Use stage							End-of-life stage				
		A1 Raw material	A2 Transport	A3 manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	
	PM (disease incidence)	1.24E-08	6.67E-10	4.33E-10	2.28E-10	3.67E-09	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	6.03E-11	3.01E-11	7.12E-12	2.33E-10
	IRHH (kg U235 eq/FU)	8.21E-03	6.39E-04	1.22E-03	2.32E-04	3.44E-03	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	5.71E-05	4.59E-05	4.80E-06	-1.16E-04
	ETF (CTUe/FU)	4.58E+00	1.17E-01	1.65E-01	4.40E-02	6.90E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.05E-02	7.50E-03	3.16E+00	1.44E-02
	HTCE (CTUh/FU)	8.60E-11	3.29E-12	4.28E-12	1.33E-12	1.94E-11	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.94E-13	5.77E-13	1.28E-14	1.34E-12
	HTnCE (CTUh/FU)	3.86E-09	1.27E-10	1.09E-10	4.71E-11	7.34E-10	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.14E-11	7.81E-12	3.33E-13	4.37E-11

	Land Use Related impacts (dimension less)	6.85E-01	1.01E-01	1.24E-01	3.35E-02	5.26E-01	0.00E+0 0	0.00E+00	0.00E+0 0	0.00E+00	0.00E+0 0	0.00E+00	0.00E+0 0	0.00E+00	9.01E-03	3.45E-03	2.44E-03	1.06E-02
---	---	----------	----------	----------	----------	----------	--------------	----------	--------------	----------	--------------	----------	--------------	----------	----------	----------	----------	----------

HTCE = Human Toxicity – cancer effects; HTnCE = Human Toxicity – non cancer effects; ETF = Ecotoxicity – freshwater; (potential comparative toxic unit)
 PM = Particulate Matter (Potential incidence of disease due to PM emissions);
 IRHH = Ionizing Radiation – human health effects (Potential Human exposure efficiency relative to U235);

	Global Warming Potential	The global warming potential of a gas refers to the total contribution to global warming resulting from the emission of one unit of that gas relative to one unit of the reference gas, carbon dioxide, which is assigned a value of 1. It is split up in 4: - Global Warming Potential total (GWP-total) which is the sum of GWP-fossil, GWP-biogenic and GWP-luluc - Global Warming Potential fossil fuels (GWP-fossil) : The global warming potential related to greenhouse gas (GHG) emissions to any media originating from the oxidation and/or reduction of fossil fuels by means of their transformation or degradation (e.g. combustion, digestion, landfilling, etc). - Global Warming Potential biogenic (GWP-biogenic) : The global warming potential related to carbon emissions to air (CO₂, CO and CH₄) originating from the oxidation and/or reduction of aboveground biomass by means of its transformation or degradation (e.g. combustion, digestion, composting, landfilling) and CO₂ uptake from the atmosphere through photosynthesis during biomass growth – i.e. corresponding to the carbon content of products, biofuels or above ground plant residues such as litter and dead wood.¹ - Global Warming Potential land use and land use change (GWP-luluc): The global warming potential related to carbon uptakes and emissions (CO₂, CO and CH₄) originating from carbon stock changes caused by land use change and land use. This sub-category includes biogenic carbon exchanges from deforestation, road construction or other soil activities (including soil carbon emissions).
	Ozone Depletion	Destruction of the stratospheric ozone layer which shields the earth from ultraviolet radiation harmful to life. This destruction of ozone is caused by the breakdown of certain chlorine and/or bromine containing compounds (chlorofluorocarbons or halons), Which break down when they reach the stratosphere and then catalytically destroy ozone molecules.
	Acidification potential	Acid depositions have negative impacts on natural ecosystems and the man-made environment incl. buildings. The main sources for emissions of acidifying substances are agriculture and fossil fuel combustion used for electricity production, heating and transport.
	Eutrophication potential	The potential to cause over-fertilization of water and soil, which can result in increased growth of biomass and following adverse effects. It is split up in 3: - Eutrophication potential – freshwater: The potential to cause over-fertilization of freshwater, which can result in increased growth of biomass and following adverse effects. - Eutrophication potential – marine: The potential to cause over-fertilization of marine water, which can result in increased growth of biomass and following adverse effects. - Eutrophication potential – terrestrial: The potential to cause over-fertilization of soil, which can result in increased growth of biomass and following adverse effects.
	Photochemical ozone creation	Chemical reactions brought about by the light energy of the sun creating photochemical smog. The reaction of nitrogen oxides with hydrocarbons in the presence of sunlight to form ozone is an example of a photochemical reaction.
	Abiotic depletion potential for non-fossil resources	Consumption of non-renewable resources, thereby lowering their availability for future generations. Expressed in comparison to Antimony (Sb). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.
	Abiotic depletion potential for fossil resources	Measure for the depletion of fossil fuels such as oil, natural gas, and coal. The stock of the fossil fuels is formed by the total amount of fossil fuels, expressed in Megajoules (MJ). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.
	Ecotoxicity for aquatic fresh water	The impacts of chemical substances on ecosystems (freshwater). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.
	Human toxicity (carcinogenic effects)	The impacts of chemical substances on human health via three parts of the environment: air, soil and water.

¹ Carbon exchanges from native forests shall be modelled under GWP - luluc (including connected soil emissions, derived products or residues), while their CO₂ uptake is excluded.

		<i>The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.</i>
	Human toxicity (non-carcinogenic effects)	<i>The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.</i>
	Particulate matter	<i>Accounts for the adverse health effects on human health caused by emissions of Particulate Matter (PM) and its precursors (NOx, SOx, NH3)</i>
	Resource depletion (water)	<i>Accounts for water use related to local scarcity of water as freshwater is a scarce resource in some regions, while in others it is not.</i> <i>The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.</i>
	Ionizing radiation - human health effects	<i>This impact category deals mainly with the eventual impact on human health of low dose ionizing radiation of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator.</i>
	Land use related impacts	<i>The indicator is the “soil quality index” which is the result of an aggregation of following four aspects:</i> - <i>Biotic production</i> - <i>Erosion resistance</i> - <i>Mechanical filtration</i> - <i>Groundwater</i> <i>The aggregation is done based on a JRC model. The four aspects are quantified through the LANCA model for land use.</i> <i>The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.</i>

DETAILS OF THE UNDERLYING SCENARIOS USED TO CALCULATE THE IMPACTS

A1 – RAW MATERIAL SUPPLY

This module takes into account the extraction and processing of all raw materials and energy which occur upstream to the studied manufacturing process of Dragocryl Velouté sup airless paint.

A2 – TRANSPORT TO THE MANUFACTURER

The raw materials are transported to Dragopaint manufacturing site by truck.

A3 – MANUFACTURING

This module takes into account the production process, which is composed by the mix of raw materials, using Belgian electricity mix and water for this purpose. Production losses represent 2.00%, and it is treated as inert waste landfilling. The product is packed with buckets, lids, interlayers and wooden pallets, which are then surrounded by plastic film.

A4 – TRANSPORT TO THE BUILDING SITE

Fuel type and consumption of vehicle or vehicle type used for transport	Transport, freight, lorry 16-32 metric ton, EURO5 {RER} transport, freight, lorry 16-32 metric ton, EURO5 Cut-off, U
Distance (km)	75
Capacity utilisation (including empty returns)	Default values from Ecoinvent 3.6
Volume capacity utilisation factor	Default values from Ecoinvent 3.6

The scenario modeled in this stage corresponds to specific data given by Dragopaint for the Belgian intermediate supplier. The Belgian default scenario has been considered for the transport between the intermediate supplier and the final customer.

A5 – INSTALLATION IN THE BUILDING

At the construction site, packaging materials are released. Also 2% material losses have been taken into account.

Parts of the installation	quantity	Description
Processes necessary for the installation of the product	-	Not applicable
Application materials	1.2E-3 kWh/FU	Airless machine
Material losses	2 %	Material losses residue in the buckets.
Packaging	1.60E-4 kg/FU 8.73E-5 kg/FU 3.75E-3 kg/FU 7.06E-4 kg/FU 5.91E-3 kg/FU	Interlayer (cardboard) Plastic film (polyethylene) Bucket (polypropylene) Lids (polypropylene) Wooden pallets

End of life	Recycling	Landfilling	Incineration	Reuse
Interlayer	95%		5%	
Plastic film	35%	5%	60%	
Buckets	35%	5%	60%	
Lids	35%	5%	60%	
Wooden pallet	40%		40%	20%
Material losses		100%		
Application materials		100%		

B – USE STAGE (EXCLUDING POTENTIAL SAVINGS)

If installed according to the manufacturer (DTU 59.1), Dragocryl Velouté sup airless paint does not need any maintenance, repair, replacement or refurbishment during the lifespan of 10 years.

C: END OF LIFE

The default scenario from NBN/DTD B 08-001:2017 was taken into consideration.

C1: No process is allocated to dismantle the paint itself. Therefore, no impacts are considered in this module.

C2: EoL transport follows the default scenario from NBN/DTD B 08-001:2017

C3: There is no waste treatment

C4: 5% of the paint is landfilled as inert waste, according to default scenario from NBN/DTD B 08-001:2017.

Module C2 – Transport to waste processing					
Type of vehicle	Waste transport	Distance (km)	Capacity utilisation (%)	Density of products (kg/m³)	Assumptions
Transport, freight, lorry 16-32 metric ton, euro5 {RER} mark et for transport, freight, lorry 16-32 metric ton, EURO5 Cut-off, U	Construction site to sorting plant	30	Ecoinvent 3.6		
Transport, freight, lorry 16-32 metric ton, euro5 {RER} mark et for transport, freight, lorry 16-32 metric ton, EURO5 Cut-off, U	Sorting plant to landfill	50	Ecoinvent 3.6		

End-of-life modules – C3 and C4		
Parameter	Unit	Value
Wastes collected separately	kg	0.152
Wastes collected as mixed construction waste	kg	0.008
Waste for re-use	kg	0
Waste for recycling	kg	0.152
Waste for energy recovery	kg	0
Waste for final disposal	kg	0.008

D – BENEFITS AND LOADS BEYOND THE SYSTEM BOUNDARIES

Module D is calculated, excluding any benefits or loads of allocated co-products. The energy recovery from the incineration and the recycling of the packaging material and packaging losses are considered as benefits and loads beyond the system boundaries, calculated in module D. Incineration benefits regard electricity and heat generated and are calculated according to PEF OEF method. Heating and electricity efficiencies are equal to 25.56% and 13%, respectively. Recycling impacts regard 95% of the final product itself is calculated in module D.

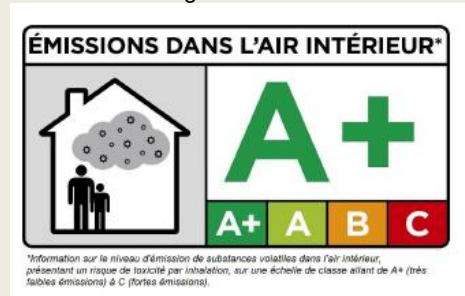
Quantitative description of the loads beyond the system boundaries	Loads represent the recycling transport done by truck 16-32 metric ton, euro5, within a 150 km, as well as the recycling processes for packaging and the final product.
Quantitative description of the benefits beyond the system boundaries	Benefits represent the recycled packaging materials as well as electricity and heat generated due to packaging incineration.

ADDITIONAL INFORMATION ON RELEASE OF DANGEROUS SUBSTANCES TO INDOOR AIR, SOIL AND WATER DURING THE USE STAGE

INDOOR AIR

This material is not classified as dangerous by the European legislation. There is no dangerous substance, according to CE n° 1272/2008. The product does not need to be stamped, according to the CE directives (67/548/ECC or 1999/45/EC).

A test following the ISO 16000 has been realized. The emission class the most beneficial is retained for this B-EPD.



SOIL AND WATER

Not applicable as this product is not in contact with water and soil.

DEMONSTRATION OF VERIFICATION

EN 15804+A2 serves as the core PCR	
Independent verification of the environmental declaration and data according to standard EN ISO 14025:2010	
Internal ☐	External ☒
Third party verifier: Evert Vermaut Vincotte Jan Olieslagerslaan 35 B-1800 Vilvoorde evermaut@vincotte.be February 2021	

APPLICATION UNIT

Results of this B-EPD are given per square meter of applied Dragocryl paint. To obtain impact results associated to the weight or the volume applied, please use the conversion factors below, defined according to the paint density and its coverage.
(only mandatory for B-EPD for use in TOTEM)

	Results for 1 m ² applied in the substrate	Results for 1 kg of Dragocryl Velouté sup airless	Results for 1 liter of Dragocryl Velouté sup airless
Conversion factor	1	6.25	8.81

- DTU 59.1 - Travaux de bâtiment - Revêtements de peinture en feuil mince, semi-épais, ou épais.
- Dragopaint Environmental Product Declaration Background Report – Decorative paints version 2.0
- ISO 14040:2006: Environmental Management-Life Cycle Assessment-Principles and framework.
- ISO 14044:2006: Environmental Management-Life Cycle Assessment-Requirements and guidelines.
- ISO 14025:2006: Environmental labels and Declarations-Type III Environmental Declarations-Principles and procedures.
- NBN EN 15804+A2:2019
- NBN/DTD B 08-001 (BE-PCR)
- Product Environmental Footprint Category Rules – Decorative Paints version 1.0 (April 2018)
- Product Environmental Footprint Category Rules – Thermal insulation
- TOTEM Lifespans (updated 15 October 2020)
- {insert relevant reference documents used}

General information

Owner of the EPD,
Responsible for the data, LCA and information

Dragopaint
Rue de l'Europe, 42 (Zone industrielle)
B-7080 Frameries
Belgium

For more information you can contact Camille Soissons
camille.soissons@dragopaint.be

EPD program
Program operator
Publisher of this EPD

B-EPD
Federal Public Service of Health / DG Environment
Galileelaan 5/2
1210 Brussels
Belgium
www.environmentalproductdeclarations.eu

Contact program operator

epd@environment.belgium.be

Based on following PCR documents

EN 15804+A2:2019
NBN/DTD B 08-001 and its complement
ISO 14025

PCR review conducted by

Federal Public Service of Health and Environment &
PCR Review committee

Author(s) of the LCA and EPD

WeLOOP
Pépinieres d'éco-entreprises – Base du 11/19 –
Bâtiment 1
Rue Léon Blum
62750 Loos-en-Gohelle
France

Identification of the project report

Dragopaint Environmental Product Declaration
Background Report – Decorative Paints version 2.0

Verification

External independent verification of the declaration and data
according to EN ISO 14025 and relevant PCR documents

Name of the third party verifier
Date of verification

Evert Vermaut, Vinçotte
01.03.2021

www.b-epd.be

www.environmentalproductdeclarations.eu

*Comparing EPDs is not possible unless they are conform to the same PCR and taking into account the building context.
The program operator cannot be held responsible for the information supplied by the owner of the EPD nor LCA practitioner.*



LCA practitioner



Building calculator of the
regional authorities

www.totem-building.be



Federal Public Service of Health,
Food Chain Safety and
Environment

www.b-epd.be